

洛伦兹力 $\vec{F}_m = q\vec{v} \times \vec{B}$

安培力 $\vec{F} = \int I d\vec{l} \times \vec{B}$

推论①两平行长导线相互作用力(单位长度受力)

$$f_{12} = f_{21} = \frac{\mu_0 I_1 I_2}{2\pi a}$$

电流同向 $\rightarrow$ 吸引力

电流反向 $\rightarrow$ 排斥力

$$\textcircled{2} \vec{M} = \vec{p}_m \times \vec{B}$$

$$\textcircled{3} \text{半圆柱面电流对轴线上长直载流导线作用力 } F = \frac{\mu_0 I_1 I_2}{\pi^2 R}$$

$$\rightarrow \text{关键: } dF = \frac{\mu_0 I_1}{2\pi R} I_2 dl = \frac{\mu_0 I_1}{2(\pi R)} I_2 dl$$

磁化强度 $\vec{M} = \lim_{V \rightarrow 0} \frac{\sum \vec{p}_m}{\Delta V}$

$|\vec{M}| = i'$ 磁化面电流密度

$$\oint_L \vec{M} \cdot d\vec{l} = \sum I_i'$$

磁场强度 $\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$ $\oint_L \vec{H} \cdot d\vec{l} = \sum I_i$

介质磁化率 χ_m : $\vec{M} = \chi_m \vec{H}$ $\mu = \mu_0(1 + \chi_m) = \mu_0 \mu_r$ $\vec{M} = \frac{\chi_m}{\mu_0 \mu_r} \vec{B}$

$$\vec{B} = \mu_0(\vec{H} + \vec{M}) = \mu_0 \mu_r \vec{H}$$

一般解题思路 $\oint_L \vec{H} \cdot d\vec{l} = \sum I_i \Rightarrow \vec{H} \Rightarrow$

由 $\vec{B} = \mu_0 \mu_r \vec{H}$ 得 $\vec{B}$
由 $\vec{M} = \chi_m \vec{H} = \mu_r - 1 \vec{H}$ 得 $\vec{M} \Rightarrow$ 由 $\vec{B} = \vec{M} \times \vec{e}_n$ 得 $\vec{v}$

法拉第电磁感应定律 $\mathcal{E}_i = -\frac{d\Phi_B}{dt}$

$$q = \frac{\Delta \Phi}{R} \text{ (画-画)}$$

动生电动势 $\mathcal{E}_i = \int_L \vec{v} \times \vec{B} \cdot d\vec{l}$

感生电动势 $\oint_L \vec{E}_i \cdot d\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$ (楞次定律)

互感系数 $M_{12} = M_{21} = \frac{\Phi_{12}}{I_1} = \frac{\Phi_{21}}{I_2}$

互感电动势 $\mathcal{E}_{12} = -M \frac{dI_1}{dt}$ $\mathcal{E}_{21} = -M \frac{dI_2}{dt}$ (楞次定律)

自感系数 $L = \frac{\Phi}{I}$

自感电动势 $\mathcal{E}_L = -L \frac{dI}{dt}$ (楞次定律)

推论①长直螺线管的自感 $L = n^2 \mu V$

②单位长度电缆(两共轴导体长薄圆筒)电感 $L = \frac{\mu}{2\pi} \ln \frac{b}{a}$ (两种方式求①定义式②磁能导出)

RL暂态电路: 电流增长 $i = \frac{\mathcal{E}}{R}(1 - e^{-\frac{Rt}{L}})$

电流衰减 $i = \frac{\mathcal{E}}{R} e^{-\frac{Rt}{L}}$

磁场能量密度 $w_m = \frac{1}{2} \vec{B} \cdot \vec{H} = \frac{B^2}{2\mu}$

磁场能量 $W_m = \int w_m dV = \int_V \frac{1}{2} \vec{B} \cdot \vec{H} dV$

位移电流密度 $\vec{j}_D = \frac{\partial \vec{D}}{\partial t}$

位移电流 $I_D = \int_S \vec{j}_D \cdot d\vec{S}$

全电流定理 $\oint_L \vec{H} \cdot d\vec{l} = I + \int_S \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S}$ 沿任意闭合回路的积分等于穿过此回路的全电流

麦克斯韦方程组

$$\oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

$$\oint_L \vec{E} \cdot d\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

$$\oint_S \vec{B} \cdot d\vec{S} = 0$$

$$\oint_L \vec{H} \cdot d\vec{l} = I + \int_S \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S}$$

电磁波的能量密度 $w = \frac{1}{2} \vec{E} \cdot \vec{D} + \frac{1}{2} \vec{B} \cdot \vec{H} = \epsilon E^2 = \mu H^2 = \sqrt{\epsilon\mu} EH = \frac{EH}{u}$

能流密度(坡印亭矢量) $\vec{S} = w\vec{u} = \vec{E} \times \vec{H}$ $\vec{S} = \frac{1}{2} E_m H_m$

波动方程 $E_y = E_m \cos(\omega t - \frac{y}{\lambda})$ $H_z = H_m \cos(\omega t - \frac{y}{\lambda})$ $E = \mu u H$ $u = \frac{1}{\sqrt{\epsilon\mu}}$

简谐振动 $x = A \cos(\omega t + \phi)$

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}} \quad \tan \phi = -\frac{v_0}{x_0 \omega} \quad \omega_0 = \sqrt{\frac{k}{m}}$$

谐振动的能量 $E_k = \frac{1}{2} m \omega^2 A^2 \sin^2(\omega t + \phi)$ $E_p = \frac{1}{2} m \omega^2 A^2 \cos^2(\omega t + \phi)$ $E = \frac{1}{2} m \omega^2 A^2 = \frac{1}{2} k A^2$

拍频 $\nu = \nu_2 - \nu_1$

李萨如图 $\frac{\omega_x}{\omega_y} = \frac{N(x=0)}{N(y=0)} = \frac{N(\text{经过} y \text{轴})}{N(\text{经过} x \text{轴})}$

- 平面简谐波 $y = A \cos \omega(t - \frac{x}{v})$
 $\Delta \varphi = \varphi_1 - \varphi_2 = (\omega t - \frac{2\pi x_1}{\lambda} + \varphi) - (\omega t - \frac{2\pi x_2}{\lambda} + \varphi) = \frac{2\pi}{\lambda} (x_2 - x_1) = \frac{2\pi}{\lambda} \Delta x = k \Delta x$
- 波的能量 $W_p = W_k = \pm \rho \Delta V \omega^2 A^2 \sin^2 \omega(t - \frac{x}{v})$ $W = \rho \Delta V \omega^2 A^2 \sin^2 \omega(t - \frac{x}{v})$
 能量密度 $w = \rho \omega^2 A^2 \sin^2 \omega(t - \frac{x}{v})$
 能流密度 $i = wu = \rho \omega^2 A^2 \sin^2 \omega(t - \frac{x}{v})$
- 波的干涉 $y_{10} = A_1 \cos(\omega t + \varphi_{10})$ $y_{20} = A_2 \cos(\omega t + \varphi_{20})$
 $\Delta \varphi = \varphi_{20} - \varphi_{10} - \frac{2\pi}{\lambda} (r_2 - r_1)$ $A^2 = A_1^2 + A_2^2 + 2A_1 A_2 \cos \Delta \varphi$
- 驻波方程 $y = 2A \cos \frac{2\pi}{\lambda} x \cos \omega t$
- 弦线上的驻波 $L = n \frac{\lambda}{2}$
- 多普勒效应 $\nu_R = \frac{u \pm \nu_R}{u \mp \nu_S} \nu_S$ $\nu_R = \sqrt{\frac{c \pm v}{c \mp v}} \nu_S$
- 光强 $I = E^2$ $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \Delta \varphi$
- 条纹间距 $\Delta x = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$
- 光程 $L = \sum n_i d_i$
 光程差 $\delta = r_2 - r_1$
- 双缝干涉 $\delta = d \sin \theta = \begin{cases} \pm k\lambda & \text{明纹 } (k=0, 1, 2, \dots) \\ \pm (2k+1)\frac{\lambda}{2} & \text{暗纹 } (k=0, 1, 2, \dots) \end{cases}$
- 等倾干涉 $\delta = 2d\sqrt{n^2 - n_1^2 \sin^2 i} = \begin{cases} k\lambda & \text{明纹 } k=1, 2, \dots \\ (2k+1)\frac{\lambda}{2}, k=0, 1, 2, \dots & \text{暗纹} \end{cases}$
 $\Delta r_k = \frac{\lambda}{2nd \sin \theta_k}$
- 劈尖干涉 $\delta = 2nd + \frac{\lambda}{2} = \begin{cases} k\lambda & \text{明纹 } k=1, 2, \dots \\ (2k+1)\frac{\lambda}{2}, k=0, 1, 2, \dots & \text{暗纹} \end{cases}$
 $l = \frac{\lambda}{2n \sin \theta} = \frac{\lambda}{2n\theta}$
- 牛顿环 $\delta = 2nd + \frac{\lambda}{2} = \begin{cases} k\lambda & \text{明纹 } k=1, 2, \dots \\ (2k+1)\frac{\lambda}{2}, k=0, 1, 2, \dots & \text{暗纹} \end{cases}$
 $\Delta r \approx \frac{\sqrt{R\lambda}}{2n\theta}$
- 迈克耳逊干涉仪 $2\Delta d = N\lambda$
- 单缝夫琅禾费衍射光强公式 $I_\theta = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$ $\alpha = \frac{\pi a \sin \theta}{\lambda}$
 明纹条件: $a \sin \theta = \pm k\lambda, k=1, 2, \dots$ 且 $a \sin \theta \neq 0$
 暗纹条件: $a \sin \theta = \pm k\lambda, k=1, 2, \dots$
- 角宽度 (主极大): $\Delta \theta_k = \frac{\lambda}{a}$
 线宽度 (主极大): $\Delta x_k = \frac{\lambda f}{a}$
- 双缝衍射的光强分布 $I_\theta = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \beta$ $\alpha = \frac{\pi a \sin \theta}{\lambda}$ $\beta = \frac{\pi d \sin \theta}{\lambda}$
 明纹条件: $d \sin \theta = \pm k\lambda (k=0, 1, 2, \dots)$ 且 $a \sin \theta \neq k\lambda$
 暗纹条件: $d \sin \theta = (2k+1)\frac{\lambda}{2} (k=0, 1, 2, \dots)$ 或 $a \sin \theta = \pm k\lambda (k=1, 2, \dots)$
- 光栅衍射 $I_\theta = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\sin N\beta}{\sin \beta} \right)^2$ $N-1$ 个根小 ≥ 0 有 $N-2$ 个次极大 N 为缝数
 明纹条件: $d \sin \theta = \pm k\lambda (k=0, 1, 2, \dots)$ 且 $a \sin \theta \neq k\lambda$
 暗纹条件: $d \sin \theta = \pm (k + \frac{m}{N})\lambda, k=0, 1, 2, \dots, m=1, 2, \dots, N-1$ 或 $a \sin \theta = \pm k\lambda$
- 半角宽 $\Delta \theta_k = \frac{\lambda}{N d \cos \theta_k}$
 中央主极大的半角宽 $2\theta_k = \frac{\lambda}{Nd}$
- 角色散率 $D_\theta = \frac{k}{d \cos \theta_k}$
 线色散率 $D_x = \frac{kf}{d \cos \theta_k}$
- 分辨率 $R = kN$
- 布喇格公式 $2d \sin \theta = \pm k\lambda, k=0, 1, 2, \dots$
- 圆孔衍射仪器分辨率 $R = \frac{D}{1.22\lambda}$
 瑞利判据 $\theta_R = 1.22 \frac{\lambda}{D}$ 最小分辨率距离 $\Delta x = u \theta_R$
- 马吕斯定律 $I = I_0 \cos^2 \alpha$
- 布儒斯特定律 $\tan i_B = \frac{n_2}{n_1}$